

SIMATS ENGINEERING
ASSESSMENT TOOL 1
Experiments

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0702

Course Code	CSA0702	Course Title	Computer Networks
CO Assessed	CO3	Assessment	Experiments
Weightage		Total Marks	

Student Name	V.Sree Ruthin Reddy	Reg.No.	192324112
Section/Batch	AIDS-2023	Date of Submission	
Faculty Incharge	DR.R S Selvan	Assessment	Individual Submission

TOTAL MARKS	20	OBTAINED MARKS	
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COURSE OUTCOME COVERED

CO3: Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions

Assessment Tool Weightage: 40%

Code of Conduct:

I V.Sree Ruthin Reddy(192324112) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student

V. Ruthin Reddy

Course Faculty
Dr.R.Senthamil Selvan

Experiments

QUESTIONS:4

Total Marks:40

1. Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, ACK). Demonstrate the role of each flag and how they establish a reliable connection. (BL4) (10)
2. Capture a DNS query using UDP in Wireshark. Compare the UDP header size with TCP and note the absence of sequence numbers or acknowledgments. Discuss how this lightweight design impacts speed and reliability. (BL4) (10)
3. Simulate packet loss (e.g., disconnect briefly) and capture traffic in Wireshark. Observe how TCP retransmits lost packets while UDP does not. Explain why this difference makes TCP reliable but slower compared to UDP. (BL4) (10)
4. Use Wireshark to capture DNS traffic while resolving a domain name. Identify the query type (A, AAAA, MX) and the response received. Measure the time taken for resolution and explain why DNS typically uses UDP. (BL4) (10)

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

understanding of concept :

TCP (Transmission Control Protocol) is a connection oriented protocol that establishes a reliable communication channel before transmitting data. It uses a Three-way Handshake consisting of three packets: SYN, SYN-ACK, and ACK. This process synchronizes sequence numbers between the client and server and ensures both devices are ready to exchange data reliably.

Use of tool:

Wireshark successfully captured the TCP packets exchanged between the client and the server.

The packet details clearly displayed the TCP flags, sequence numbers, acknowledgement numbers, source IP address, and destination IP address.

Analysis of Results :

The first packet contains the SYN flag, indicating that the client is requesting to establish a connection. The server replies with a SYN-ACK packet, acknowledging numbers, source IP address, and destination IP address.

Result :- The TCP three-way handshake was successfully captured in Wireshark. The SYN, SYN-ACK and ACK packets were identified.

2.

Capture a DNS Query using UDP and compare with TCP.

The Domain Name System (DNS) converts human readable domain names into IP addresses. DNS generally uses the UDP protocol because it provides faster communication with minimal overhead. Unlike TCP, UDP is connectionless and does not establish a connection before sending data.

Experimental Design

1. Open Wireshark
2. Start package capture
3. Open a browser and access any website
4. Stop packet capture.
5. Apply the filter dns.
6. Observe the DNS query and UDP header information.

Wireshark captured the DNS request and response packets transmitted over UDP Port 53. The UDP header and DNS details were successfully displayed.

Analysis of Results

The UDP header size is only 8 bytes, whereas the minimum TCP header size is 20 bytes. UDP does not contain sequence numbers, acknowledgement numbers, or connection establishment mechanisms. This reduces

Communication overhead and makes UDP much faster than TCP. However, since UDP does not retransmit lost packets or acknowledge received packets.

TCP guarantees reliable data delivery by detecting lost packets and retransmitting them. UDP, on the other hand, is a connectionless protocol. Therefore, TCP is reliable but slower, whereas UDP is faster but does not guarantee delivery.

Experimental Design :

1. Open Wireshark and begin packet capture.
2. Start downloading a file or open a web page.
3. Temporarily disconnect the internet connection for a few seconds.
4. Reconnect the network.
5. Stop packet capture.
6. Apply the filter `tcp.analysis.transmission` to identify the transmitted packets.

Use of Tool :-

Wireshark successfully detected TCP retransmission packets after the temporary network interruption. UDP packets showed no transmission because UDP does not provide recovery mechanisms.

Analysis of Results

When packet loss occurred, TCP detected the missing acknowledgement and retransmitted the lost packets automatically. This ensured that all data reached the destination correctly. UDP continued transmitting without attempting to recover lost packets, which may result in missing information.

Documentation :

Wireshark confirmed that TCP retransmitted lost packets, whereas UDP did not. This demonstrates the reliability of TCP and the speed advantage of UDP.

4. Capture DNS Traffic and Analyze Query Type & Response.

DNS translates domain names into IP addresses so that computers can communicate over the Internet.

Common DNS Query types include A (IPv4 Address),

AAAA (IPv6 address), and MX (Mail Exchange).

DNS generally uses UDP because it provides fast & efficient communication.

Experimental Design :

1. open wireshark.
2. start packet capture
3. open a website

4. stop packet capture
5. Apply the filter dns.
6. observe the DNS Query type, response and resolution time.

Wireshark displayed the DNS request, response packet, query type, IP address returned, and the time taken for DNS resolution.

Analysis of Results

The captured DNS packet showed a Query Type A, requesting the IP4 address of the website. The DNS server responded with the corresponding IP address. This reduces network overhead and improves response speed.